

Unit 4 Lab Assignments

4-A: Introduction to Optimization

There is often more than one way to write a program that produces correct results. In such cases, we consider factors such as speed and memory-use to determine which program solution is best.

A college requires all students to take a math placement test. The placement score determines the math class the student must take.

Score	Placement
300 or higher	Honors Math
150-299	Algebra I
Less than 150	Remedial Math

Consider the following two examples. Both examples produce the same correct result. Which program fragment do you think is better? Why? **Submit answers in a document** in Blackboard as **Lab 4-A** in the **Lab 4 submission link**.

Fragment #1

```
if (mathScore >=300)
    cout << "Placement: Honors Math" << endl;
if ((mathScore >= 150) && (mathScore < 300))
    cout << "Placement: Algebra I" << endl;
if (mathScore < 150)
    cout << "Placement: Remedial Math" << endl;
```

Fragment #2

```
if (mathScore >= 300)
    cout << "Placement: Honors Math" << endl;
else
    if (mathScore > 150)
        cout << "Placement: Algebra I" << endl;
    else
        cout << "Placement: Remedial Math" << endl;
```

MY GUESS:

I believe Fragment #1 will be the faster one since it explicitly defines the bounds for which any condition may fall during runtime. I speculate that this will make compilation a simpler task or lead to a simpler compiled version, thus reducing calculation time.

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4-B: Coding from Documentation

Your text documents various C++ functions in Appendix F.

Refer to the functions contained in the ctype (ctype.h) library. Identify a function that will determine if a char input is a digit ('0' to '9').

Indicate how many pieces of data must be passed to the function.

Indicate what (if anything) will be returned by the function.

Demonstrate the appropriate use of the function in a **short code fragment**.

Submit your answers and short code fragment as **Lab 4-B** in the **Lab 4 submission link**.

4-C: Voter Tally

Citizens in a nearby county recently voted whether to approve a bond referendum to finance the construction of a new school. The first five results of that vote are contained in the file `votes.dat`.

Write a program that reads the file `votes.dat`. The program should summarize the voting data to include a total tally of "yes" and "no" votes as well as a break-down of votes based on the voter's district.

For example, a sample `votes.dat` file might contain the data shown below. The first value within a line is the voter's district (1-3). The second value within a line indicates whether the voter approved (Y) or disapproved (N) of the measure. Remember, you may quickly make a data file using any text editor such as Windows Notepad.

Submit your `.dat` file and your code as **Lab 4-C** in the **Lab 4 submission link**

```
1      Y
2      N
3      N
3      Y
1      Y
2      Y
```